

Methods: EDA and linear regression

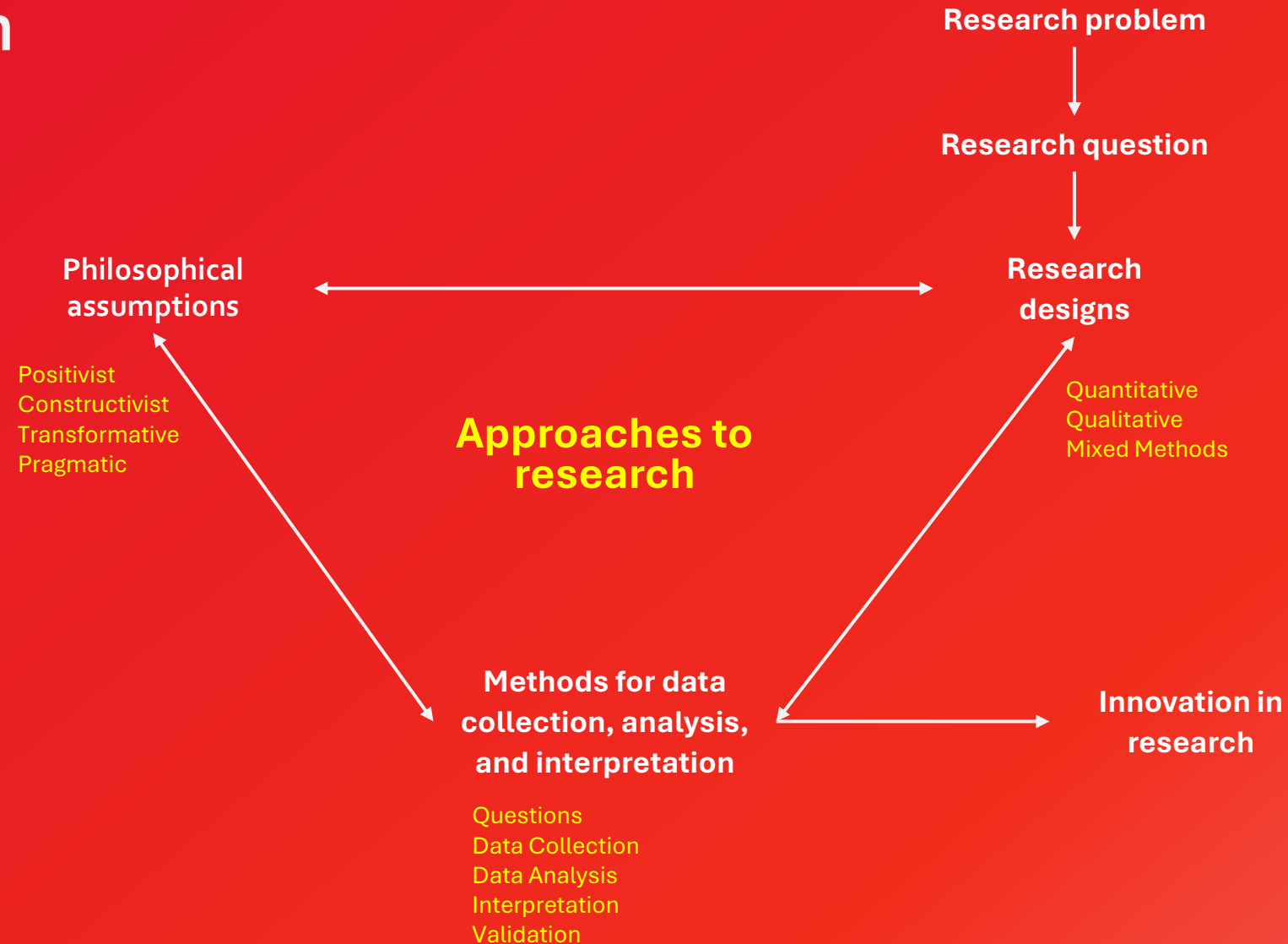


Luis Carlos Guevara

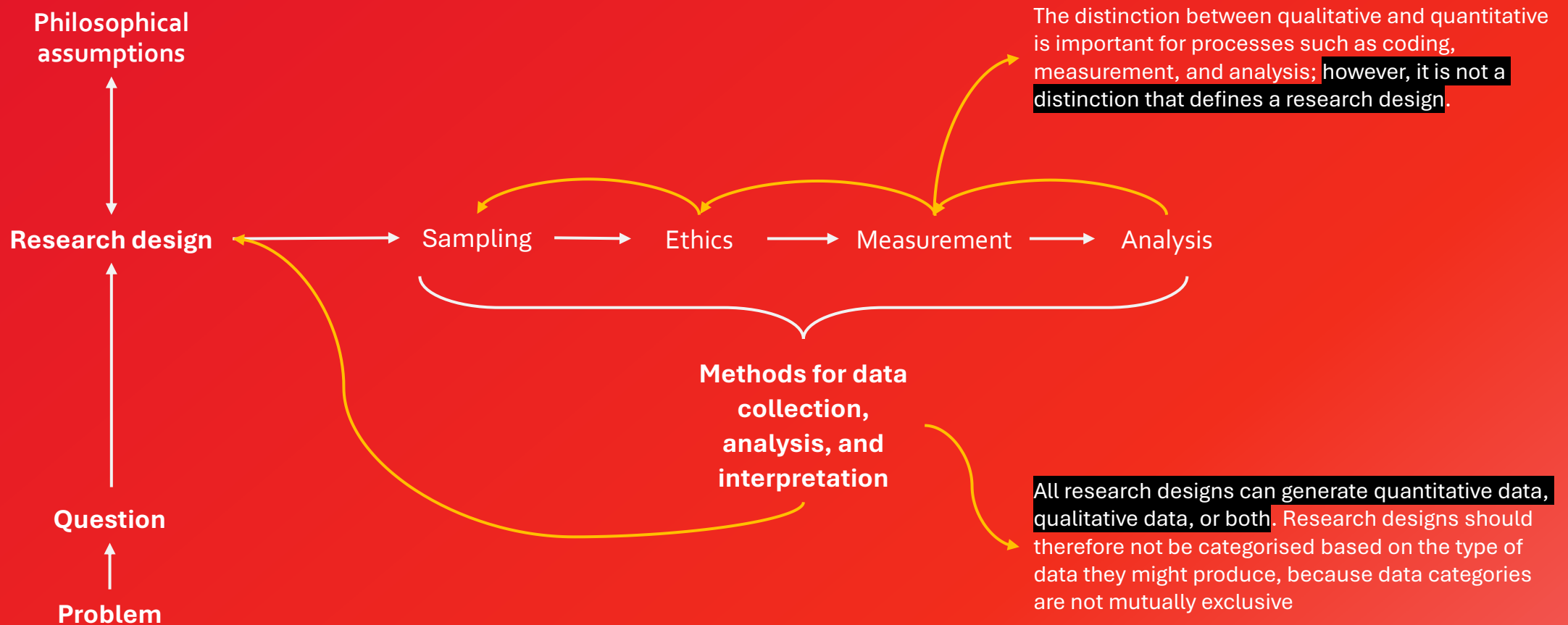
Faculty of Finance,
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2026-I

A Framework for Research



A Framework for Research, from the problem



What is the purpose of an Exploratory Data Analysis?

To examine the data in terms of its:

- Distribution
- Relationships between variables
- Outliers and anomalies

- ✓ Developing an EDA process evaluates data quality and supports hypothesis model building.
- ✓ Most EDA techniques are graphical, with only a few quantitative approaches.



[John W Tukey \(1977\)](#)

John W. Tukey (1977) introduced EDA, and since then it has become standard practice for analysing datasets.



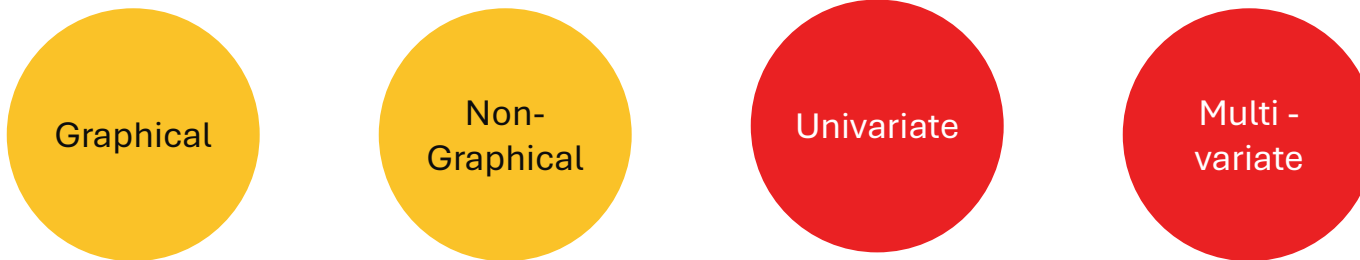
[Howard Seltman](#)

“Any method that examines the data without formal statistical modelling or inference is considered exploratory data analysis”.

EDA is a crucial step after data collection and preprocessing, where the data are visualised, plotted, and manipulated without assumptions.

What is the purpose of an Exploratory Data Analysis?

- 1.** **To maximise knowledge of the dataset,** that is, to develop a thorough understanding of its structure
- 2.** **To visualise potential relationships (direction and magnitude)** between independent and dependent variables.
- 3.** **To detect outliers and anomalies,** which are values that differ significantly from the rest of the observations.
- 4.** **To extract and create relevant variables.**
- 5.** **To develop parsimonious models,** meaning explanatory models that use the smallest number of variables possible, or to identify a preliminary selection of appropriate models



Types of Data and Graphical EDA Techniques

Data type	EDA technique
Categorical	Descriptive statistics, bar charts
Univariate continuous	Histogram, line graph
Bivariate continuous	2D scatter plot
2-D matrices	Heatmaps
Multivariate – trivariate	3D scatter plot; 2D scatter plot with categorical variables; segmented scatter plots by category; boxplots by groups

Types of Data and Graphical EDA Techniques

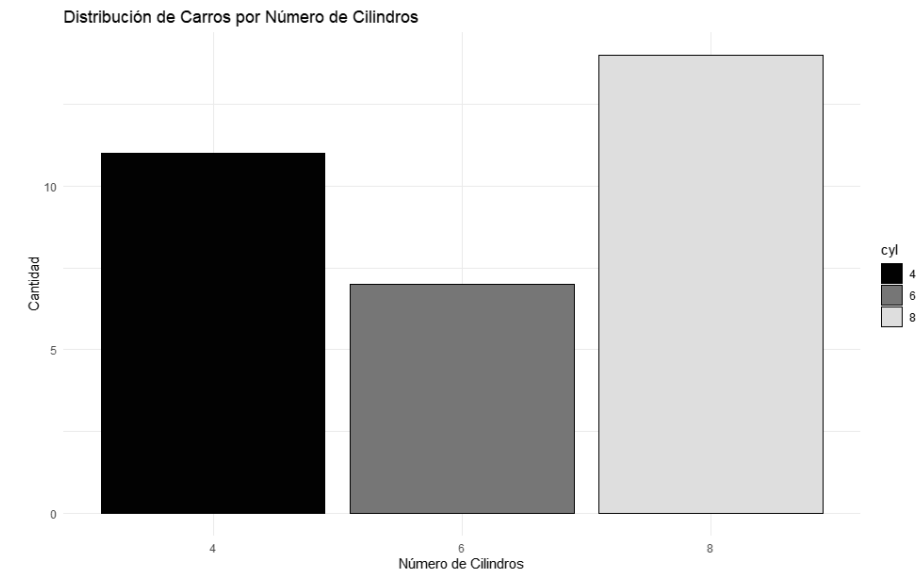
1. Categorical variables and counts

Freq
tables

```
cyl_table <- table(mtcars$cyl)
print(cyl_table)
```

Bar
plots

```
ggplot(mtcars, aes(x = cyl, fill = cyl)) +
  geom_bar(color = "black") +
  scale_fill_viridis_d(option = "magma") +
  labs(x = "Número de Cilindros",
       y = "Cantidad",
       title = "Distribución de Carros por Número de Cilindros")
+ theme_minimal()
```



Types of Data and Graphical EDA Techniques

2. Continuous univariate

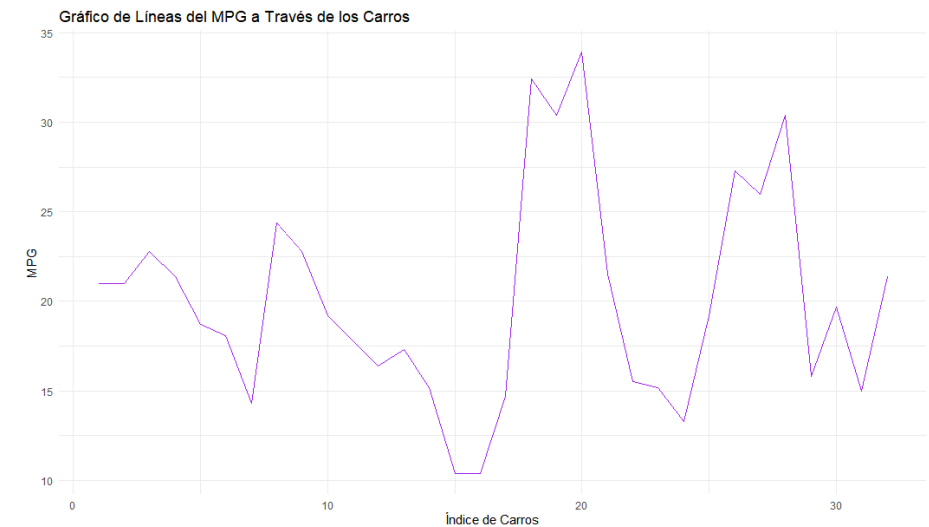
Histogram

```
ggplot(mtcars, aes(x = mpg)) +
  geom_histogram(binwidth = 2, aes(fill = ..count..), color =
    "black") + scale_fill_viridis_c(option = "magma") +
  labs(x = "Millas por Galón (mpg)",
    y = "Frecuencia",
    title = "Histograma de MPG") +
  theme_minimal()
```

Line plots

```
mtcars <- mtcars %>% mutate(index = row_number())

# Gráfico de líneas del MPG a través del índice de carros
ggplot(mtcars, aes(x = index, y = mpg)) +
  geom_line(color = "purple") +
  labs(x = "Índice de Carros",
    y = "MPG",
    title = "Gráfico de Líneas del MPG Vs Carros") +
  theme_minimal()
```



Types of Data and Graphical EDA Techniques

3. Continuous bivariate

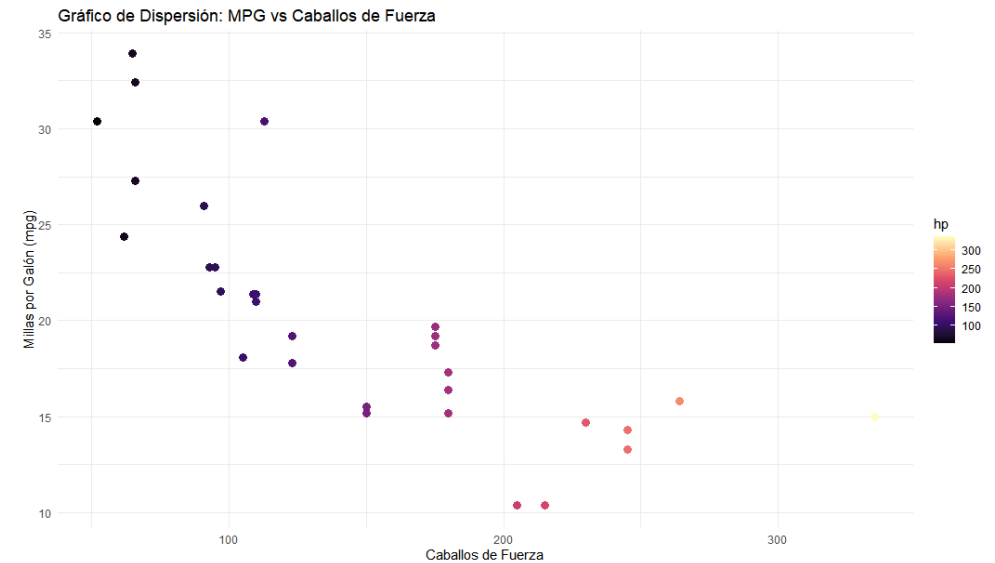
Scatter plot

```
ggplot(mtcars, aes(x = hp, y = mpg)) +
  geom_point(size = 3, aes(color = hp)) +
  scale_color_viridis_c(option = "magma") +
  labs(x = "Caballos de Fuerza",
       y = "Millas por Galón (mpg)",
       title = "Gráfico de Dispersión: MPG vs Caballos de Fuerza") +
  theme_minimal()
```

Correlation matrix

```
# Selecciona variables continuas para la correlación: MPG,
disp, HP y peso (wt)
numeric_vars <- mtcars %>% select(mpg, disp, hp, wt)
corr_matrix <- round(cor(numeric_vars), 2)
melted_corr <- melt(corr_matrix)

# Mapa de calor usando la paleta magma
ggplot(melted_corr, aes(x = Var1, y = Var2, fill = value)) +
  geom_tile(color = "white") +
  scale_fill_viridis_c(option = "magma", direction = -1) +
  labs(x = "", y = "", title = "Mapa de Calor de
Correlación para Variables Seleccionadas de mtcars") +
  theme_minimal()
```



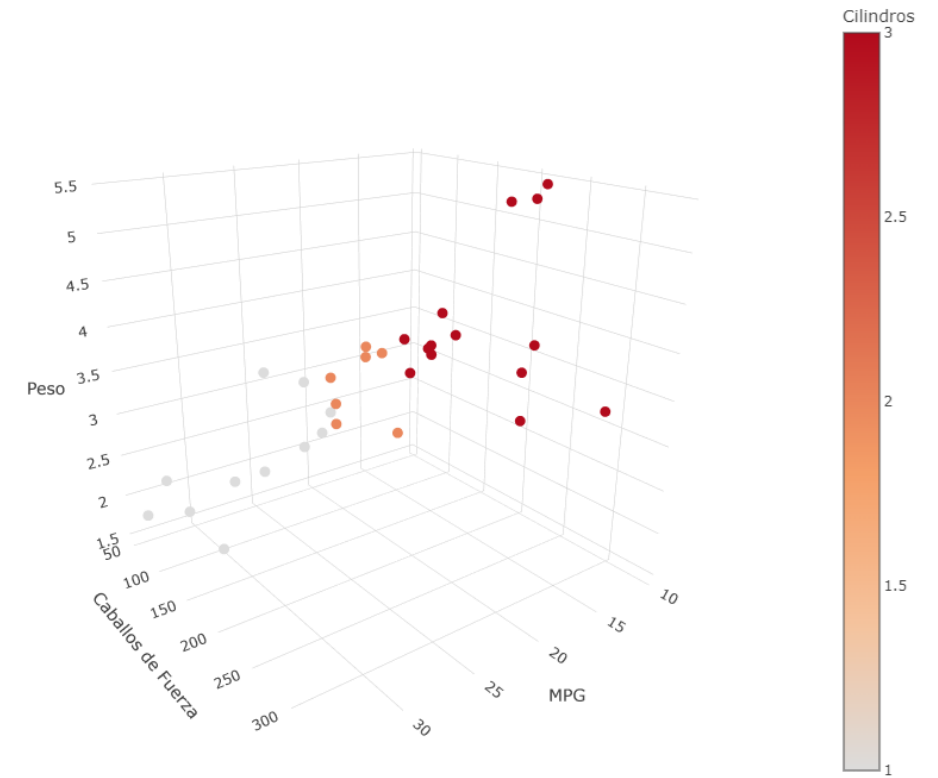
Types of Data and Graphical EDA Techniques

4. Continuous multivariate

3D Scatter plot

```
p3d <- plot_ly(mtcars,
  x = ~mpg,
  y = ~hp,
  z = ~wt,
  type = "scatter3d",
  mode = "markers",
  marker = list(size = 5, color = ~as.numeric(cyl),
    colorscale = "Magma",
    colorbar = list(title = "Cilindros")) %>%
  layout(title = "Gráfico 3D: MPG vs Caballos de Fuerza vs Peso",
    scene = list(xaxis = list(title = "MPG"),
      yaxis = list(title = "Caballos de Fuerza"),
      zaxis = list(title = "Peso")))
```

Gráfico 3D: MPG vs Caballos de Fuerza vs Peso



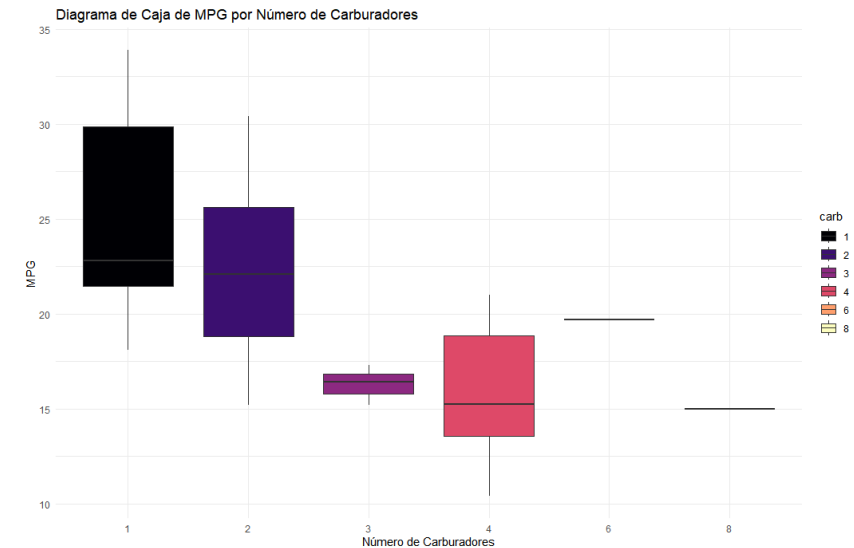
Types of Data and Graphical EDA Techniques

4. Continuous multivariate

Box plots

```
ggplot(mtcars, aes(x = carb, y = mpg, fill = carb)) +
  geom_boxplot() +
  scale_fill_viridis_d(option = "magma") +
  labs(x = "Número de Carburadores",
       y = "MPG",
       title = "Diagrama de Caja de MPG por Número de Carburadores") + theme_minimal()

ggplot(mtcars, aes(x = carb, y = index, fill = carb)) +
  geom_boxplot() +
  scale_fill_viridis_d(option = "magma") +
  labs(x = "Número de Carburadores",
       y = "MPG",
       title = "Diagrama de Caja de MPG por Número de Carburadores") + theme_minimal()
```



Types of Data and non-graphical EDA Techniques

Given an X numerical variable

Function	Description
summary(x)	General description of a vector
max(x)	Max value
mean(x)	Average value
median(x)	Median value
min(x)	Min Value
sd(x)	Standard deviation
var(x)	Variance
IQR(x)	Interquartile range

Types of Data and non-graphical EDA Techniques

1. Continuous univariate

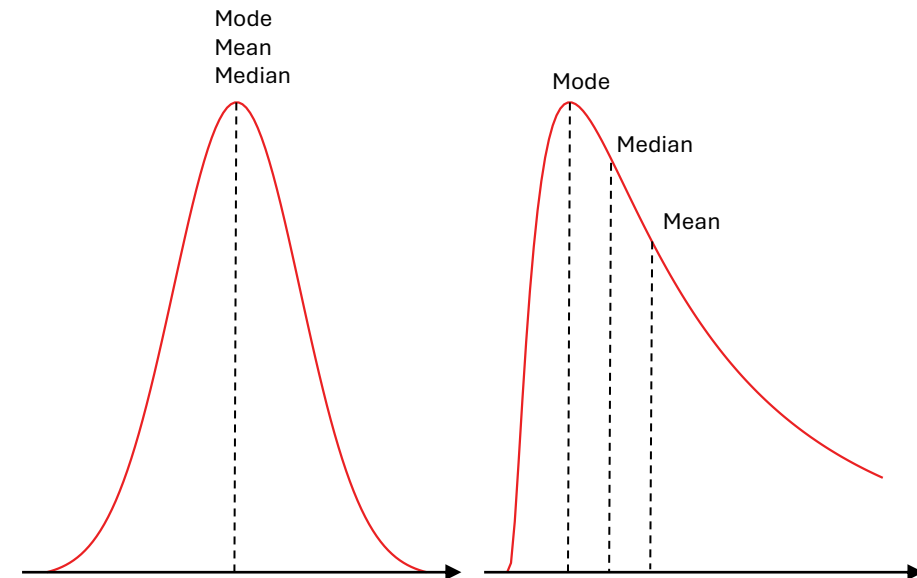
Frequency tabulation for categories.

Used for categorical variables. It consists of constructing a table that contains the count and the proportion (or frequency) of the data corresponding to each category.

Central tendency, dispersion, and distributional shape (skewness, kurtosis). These rely on sample statistics to **express the characteristics of a sample using a limited set of parameters**, which serve as estimates of the corresponding population parameters.

- ✓ **Measures of central tendency** (arithmetic mean, median, mode)
- ✓ **Measures of dispersion** (variance, standard deviation, interquartile range, minimum and maximum values)
- ✓ **Measures of distribution** (skewness, kurtosis)

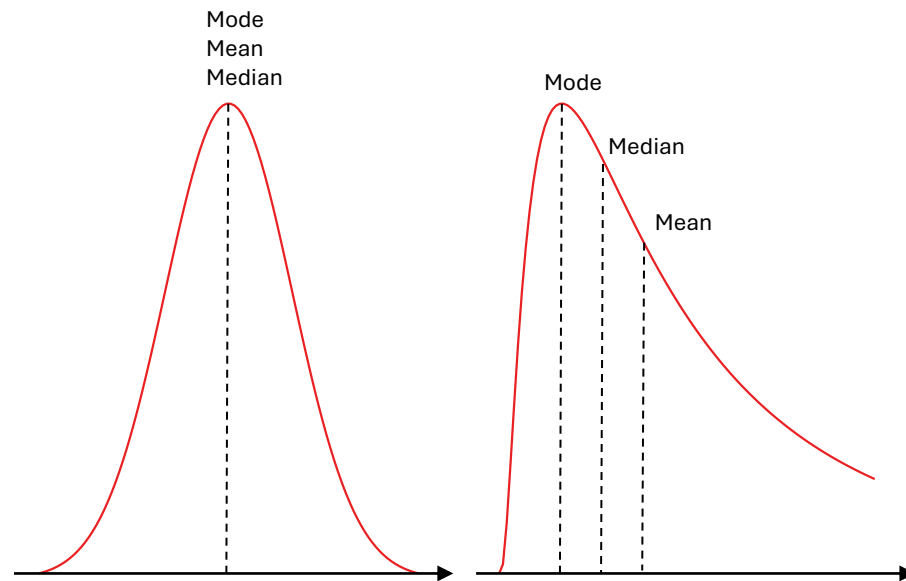
The median is the central value in an ordered list of data. **Because the median is less affected by extreme values and outliers**, it is considered more “robust” than the mean.



Types of Data and non-graphical EDA Techniques

Measures of Central Tendency

The **arithmetic mean** is the sum of all observations divided by the number of values. **The median** is the central value in a list containing all observations ordered from smallest to largest.



Sample mean

$$\bar{x} = \frac{\sum x_i}{n}$$

$$\bar{x} = \frac{\sum x_i}{n} = \frac{(x_1 + x_2 + \dots + x_n)}{n}$$

Sample median

To determine the median of a sample with n observations:

- Order the data from smallest to largest.
- If n is odd, the median is the central observation.
- If n is even, the median is the mean of the two central observations.

$$\frac{(y_{[n/2]} + y_{[n/2+1]})}{2}$$

Types of Data and non-graphical EDA Techniques

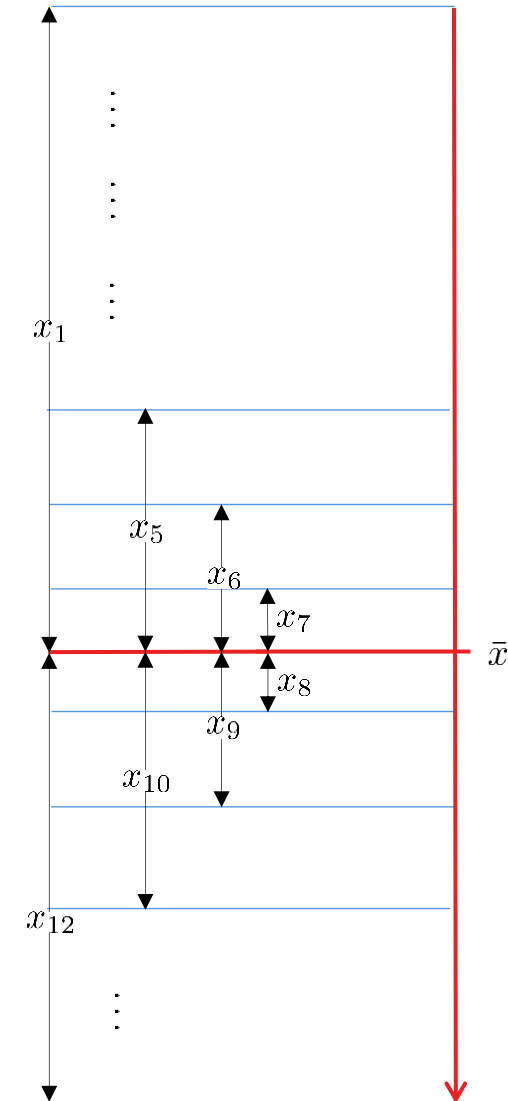
Variance

When variance is calculated using the entire population, the result is obtained by dividing the sum of squares by the population size.

The sample formula for variance conventionally uses $n-1$ in the denominator to achieve the property of unbiasedness.

This means, approximately, that when calculated for many different random samples from the same population, **the average of the sample variances should coincide with the corresponding population parameter.**

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{(n-1)}$$



Types of Data and non-graphical EDA Techniques

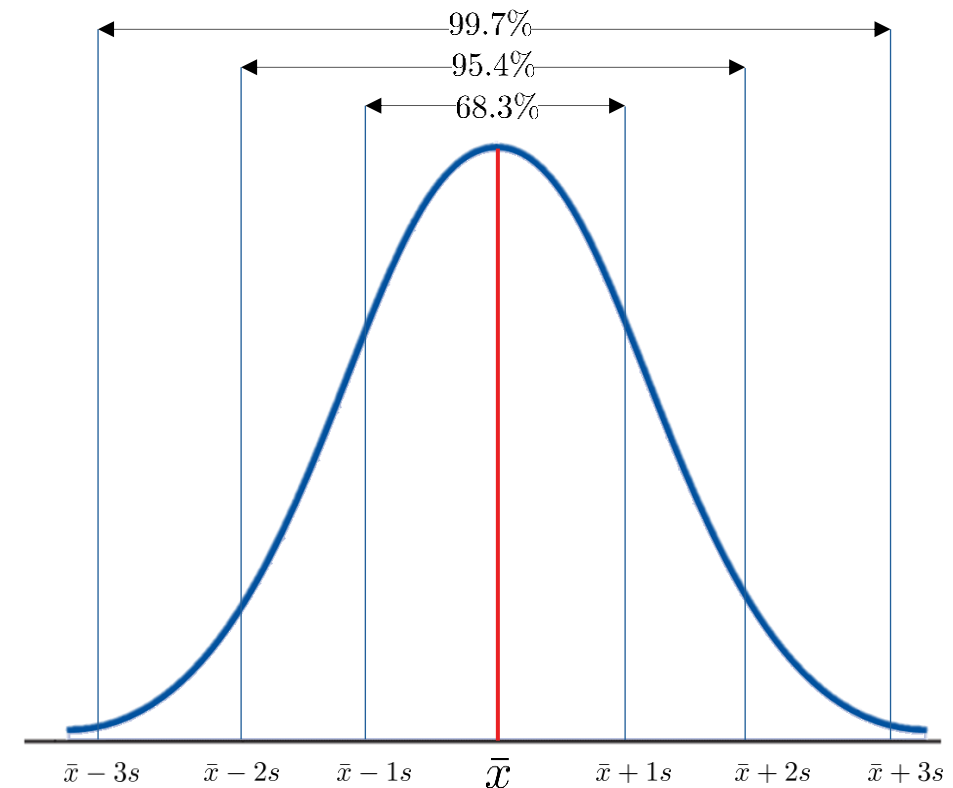
Standard Deviation

The standard deviation is the square root of the variance. Because of this, it has the same units as the original data, which makes it more interpretable.

The sample standard deviation is commonly represented by the symbol s .

For a theoretical Gaussian distribution, the mean $\pm 1, 2$, or 3 standard deviations encompasses 68.3%, 95.4%, and 99.7% of the probability density, respectively. This is known as the empirical interval.

$$s = \sqrt{s^2} = \sqrt{\frac{\sum (x_i - \bar{x})^2}{(n-1)}}$$



Types of Data and non-graphical EDA Techniques

Interquartile Range (IQR)

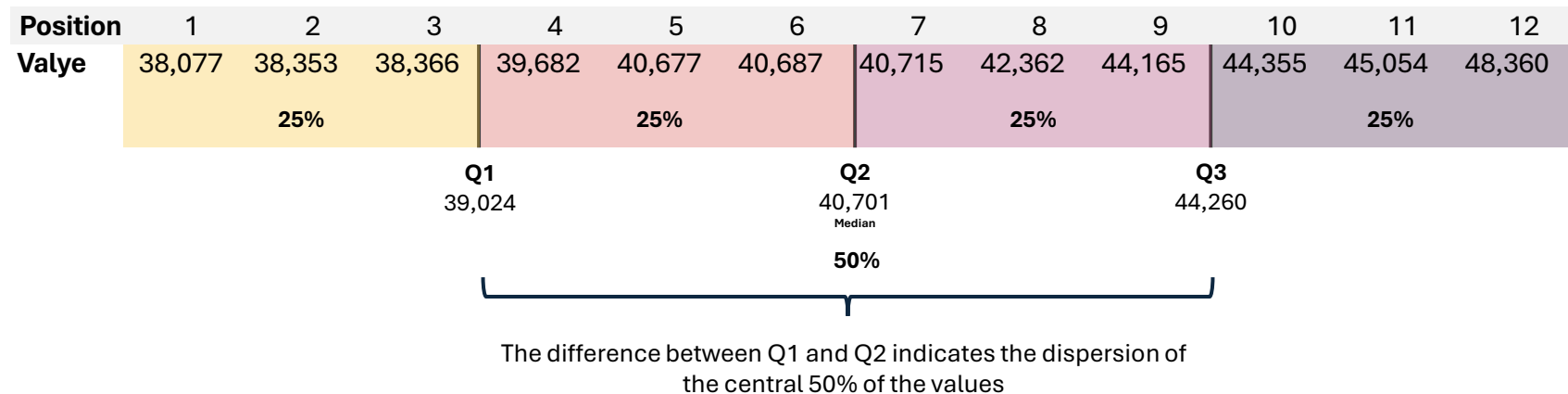
The interquartile range (IQR) is calculated using the limits of the data located between the first and third quartiles. **The IQR is a more robust measure of dispersion than the variance or the standard deviation and should therefore be preferred for small or asymmetric distributions.**

IQR

$$Q1 = \left(\frac{p=25}{100}\right)n \quad Q3 = \left(\frac{p=75}{100}\right)n$$

$$iqr = Q_3 - Q_1$$

- **For symmetric distributions** (not necessarily normal) and $N > 30$, results should be expressed as mean $\pm$ standard deviation.
- **For asymmetric distributions**, $N < 30$, or when there is evidence of outliers, results should be expressed as median $\pm$ IQR, as these are more robust.



Types of Data and non-graphical EDA Techniques

Cross tabulation

Cross-tabulation represents a bivariate non-graphical exploratory data analysis (EDA) technique. **It is an extension of tabulation and is applicable to categorical data as well as quantitative data with only a few variables.**

For two variables, a two-way table is constructed with:

- **Column headers** corresponding to the levels of one variable, and
- **Row headers** corresponding to the levels of the other variable.

The table is then filled with the counts of all individuals who share each pair of levels.

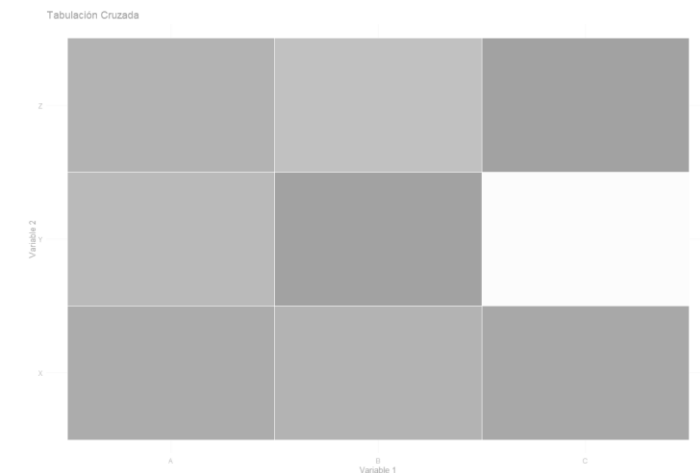
Cross tabulation

```
# Crear tabla de tabulación
cruzadatabla_cruzada ← table(datos$Variable1, datos$Variable2)

# Convertir tabla cruzada a un data frame
tabla_cruzada_df ← as.data.frame(tabla_cruzada)

# Agregar columna de frecuencia relativa
tabla_cruzada_df$FrecuenciaRelativa ← tabla_cruzada_df$Freq /
sum(tabla_cruzada_df$Freq)
```

Var1	Var2	Freq	Relative frequency
A	X	10	0.1
B	X	11	0.11
C	X	9	0.09
A	Y	12	0.12
B	Y	8	0.08
C	Y	18	0.18
A	Z	11	0.11
B	Z	13	0.13
C	Z	8	0.08



Types of Data and non-graphical EDA Techniques

Covariance and Correlation

These measures assess the degree of relationship between two random variables and express the extent to which they vary together.

Covariance

$$1. \quad \bar{x} = \frac{\sum x_i}{n} \quad \Bigg| \quad \bar{y} = \frac{\sum y_i}{n}$$

$$2. \quad (x_i - \bar{x}) \quad \Bigg| \quad (y_i - \bar{y})$$

$$3. \quad (x_i - \bar{x})(y_i - \bar{y})$$

$$4. \quad S_{xy} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{n-1}$$

Correlation coefficient

$$r_{xy} = \frac{s_{xy}}{s_x \times s_y}$$

The correlation test helps determine the degree of association between two variables.

This association does not imply causality. The correlation coefficient is a number between -1 and 1 that indicates the strength and direction of the relationship between variables:

- ✓ **1 indicates a perfect positive correlation:** as one variable increases, so does the other.
- ✓ **0 indicates no correlation:** the variables show no linear association.
- ✓ **-1 indicates a perfect negative correlation:** as one variable increases, the other decreases.

Types of Data and non-graphical EDA Techniques

Pearson Correlation Coefficient

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}$$

The Pearson correlation measures the linear relationship between two continuous quantitative variables. It evaluates how the values of both variables vary together while maintaining a proportional relationship. The coefficient ranges from -1 to 1:

- Pearson's correlation is sensitive to outliers, as these can considerably influence its value.

Spearman Correlation Coefficient

$$\rho = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)}$$

Where: (d_i) is the difference between the ranges of (X) and (Y) , calculada described as: $(R(x_i) - R(y_i))$.

The Spearman correlation evaluates the monotonic relationship between two variables by ranking their values.

- SCC is non-parametric, meaning it does not require assumptions about the distribution of the data.
- This coefficient is useful when relationships are not strictly linear but follow a consistently increasing or decreasing pattern.
- As with Pearson's coefficient, its values range from -1 to 1, but Spearman is more robust in the presence of outliers and can be applied to ordinal data.

Identification of Data Types and Study Objectives

Study Objectives

Identifying the objective of the study is an extremely important aspect when planning data analysis. **A vague or poorly described objective often leads to poorly executed analysis.**

The study objective must clearly identify:

- The study population
- The outcome of interest
- The covariate(s) of interest
- The relevant temporal points of the study
- What is intended to be done with these elements

Investing time in making the objective as specific and clear as possible often saves time in the long term.

- ✗ “To evaluate the reduction in the unemployment rate within the population that has completed tertiary education in Colombia over the past five years.”
- ✓ “To evaluate the impact of programmes aimed at expanding access to higher education in rural areas of Colombia, considering as the study population young people aged 17–25 from municipalities with low educational access. The outcome of interest will be the increase in enrolment and graduation rates, while covariates will include available educational infrastructure, socioeconomic status, and governmental support received. The evolution of these factors will be analysed over the past seven years.”

Data types

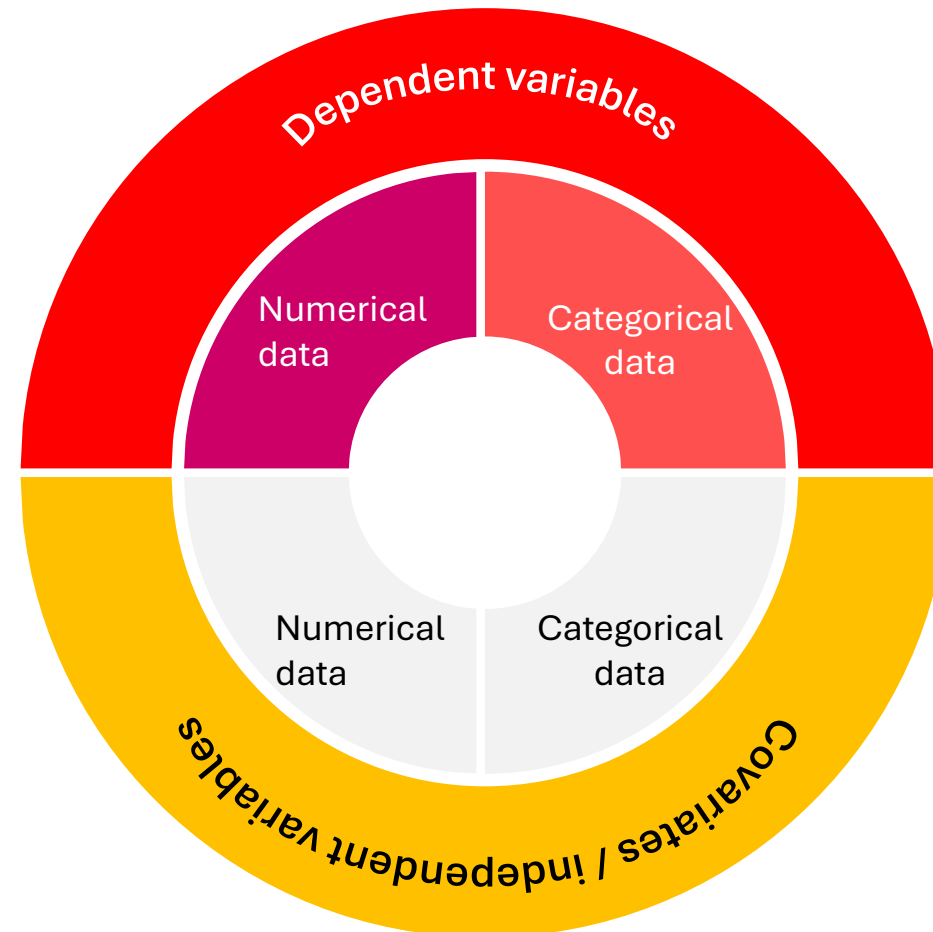
After specifying a clear study objective, the next step is to **determine the types of data that will be used in the analysis.**

Numerical data are values that may take any numerical value within a range.

- For example, graduates' income levels would be continuous data in a study analysing the impact of education on employment.

Categorical data are values that take a limited number of categories.

- Such as educational level (technical, undergraduate, postgraduate).



Outcomes are the variables that the study aims to investigate, improve, or influence.

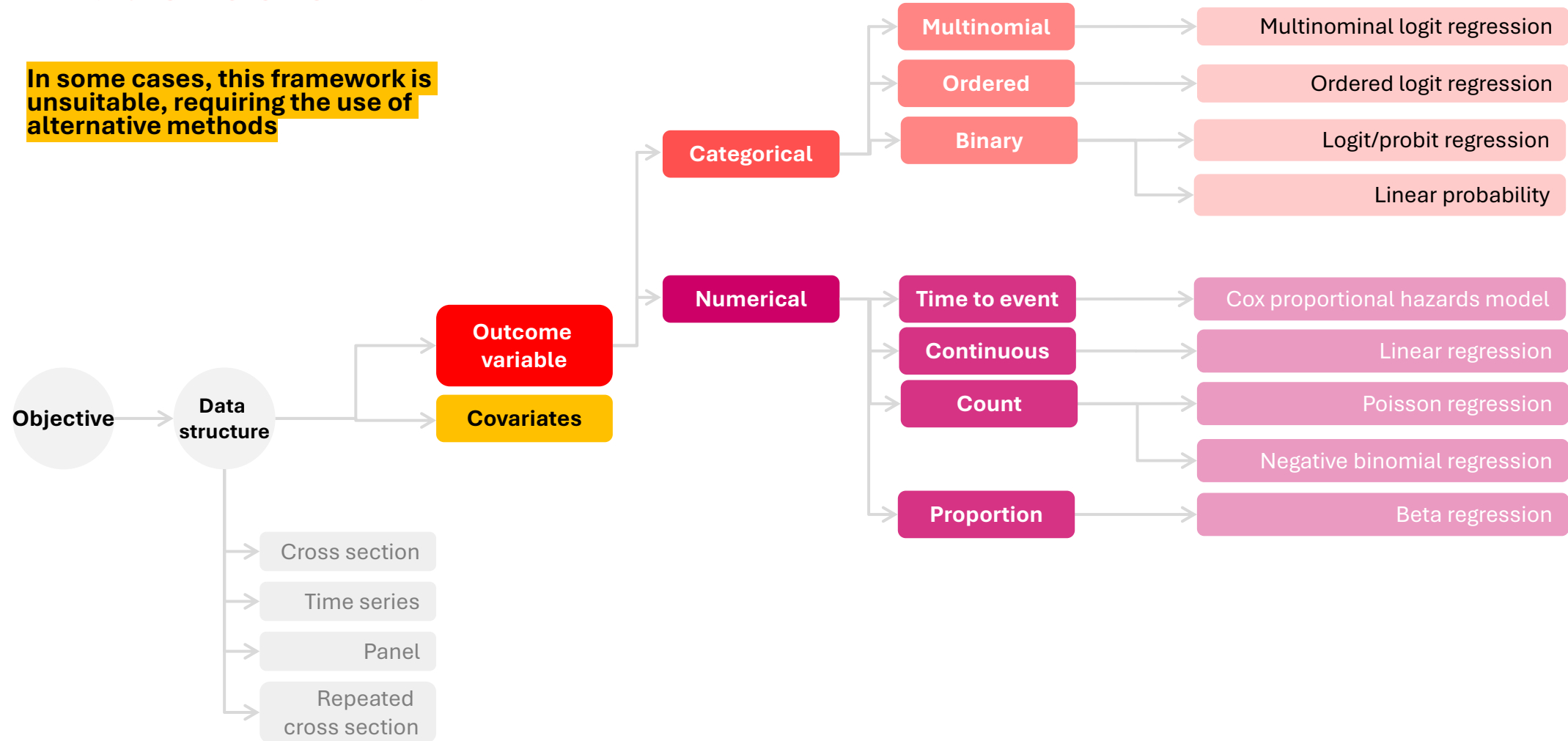
- In the case of analyzing the impact of tertiary education on employability, the outcome would be the rate of labor market insertion

Covariates are variables that may influence the outcome or should be considered to avoid bias.

- For instance, in assessing the impact of tertiary education on employment, covariates may include educational level attained, field of study, and region of residence.

Data types and model selection

In some cases, this framework is unsuitable, requiring the use of alternative methods



Linear regression



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2026-I

Linear regression

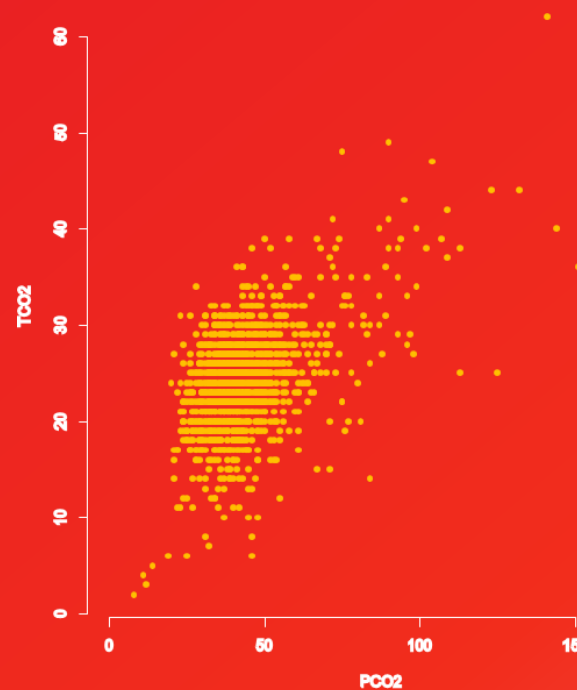
Linear regression provides the foundation for many analytical approaches used to study a dependent continuous variable in relation to a set of independent variables.

In the simplest case (univariate linear regression) the objective is to relate a continuous outcome to a single continuous or categorical covariate, by estimating values for β_0 and β_1 in the model:

$$y = \beta_0 + \beta_1 x$$

such that the equation fits the observed data in an optimal way. The estimates ($\hat{\beta}_0$) y ($\hat{\beta}_1$) are distinguished from the true (often unknown) parameter values.

Scatter Plot: PCO₂ and TCO₂ in a Sample of 1,776 Patients



A scatter plot of total carbon dioxide (TCO₂) (dependent variable, y-axis) against partial pressure of carbon dioxide (PCO₂) (independent variable, x-axis) shows that as PCO₂ increases, TCO₂ also increases.

This visual pattern suggests that a linear regression model could be appropriately fitted to describe how TCO₂ varies as a function of PCO₂.

Simple linear regression

Simple linear regression **is used to examine the relationship between a single independent variable and a continuous dependent variable**, assuming that this relationship can be adequately represented by a straight line.

“Explain y in terms of x , or study how y changes when x changes”

- ✓ Are there exact relationships between x and y ?
- ✓ Is there a **functional relationship** between x and y ?
- ✓ Is there a **ceteris paribus** relationship between x and y ?

Simple linear regression

$$y = \underbrace{\beta_0 + \beta_1 x}_{\text{Systematic part}} + \underbrace{u}_{\text{Non-systematic part}}$$

Systematic part of the model: the component of y that is explained by x .

Non-systematic part of the model: the component of y that is explained by u .

Simple linear regression model Or bivariate linear regression model

This model is assumed to hold in the population of interest, and it enables us to relate two variables: a **dependent variable, an independent variable, and an error term** (or disturbance) that captures unexplained variation in the relationship.

A simple regression analysis treats all factors that affect y , and which are different from x , **as unobserved factors**.

$$\Delta y = \Delta \beta_0 + \Delta \beta_1 x + \Delta u$$

$$\Delta y = \beta_1 \Delta x$$

is the **slope parameter in the relationship between y and x** , when all other factors contained in u remain constant.

Simple linear regression

How to assess the relationship between the independent variable and the error term?

- ✓ Correlation coefficient
- ✓ Conditional expectation

U and X are random variables.

The conditional distribution of u given any value of x can be defined as follows:

For each x , the expected value (or average) of u can be obtained in the subset of the population described by that value of x .

$$E[u|x] = E[u] \quad \left| \begin{array}{l} \text{Mean independence of } U \text{ given } X. \\ \text{The average value of } u \text{ does not depend on the value of } x \end{array} \right.$$

$$E[u|x] = 0 \quad \left| \begin{array}{l} \text{Conditional mean zero assumption.} \\ \text{The average value of } u \text{ does not depend on the value of } x. \end{array} \right.$$

Simple linear regression

$$Wage = \beta_0 + \beta_1 educ + u$$

The wage is measured as the pesos earned per hour, and education is measured in years: therefore, β_1 **measures the change in the hourly wage for each additional year of education, keeping all other factors constant.**

- Work experience
- Innate abilities
- Tenure in the current job
- Work ethic

According to the assumptions about the relationship between X and U, it is required, for example, that the average level of ability be the same regardless of years of schooling.

$$E[cap|x = 8] = E[cap|x = 16] = E[cap|x = 2] = \dots$$

In which cases would we expect this model to satisfy the equation $E[u|x] = 0$?

In an experimental context, for example, causal inference based on regression methods is developed under the assumption that when the observed **variables are “made equal” in the control and treatment groups**, the selection bias from factors that we cannot observe is eliminated.

Simple linear regression

Population regression function

$$y = \beta_0 + \beta_1 x + u$$

$$E[y|x] = E[\beta_0 + \beta_1 x + u|x]$$

$$E[y|x] = E[\beta_0|x] + E[\beta_1 x|x] + E[u|x]$$

$$E[y|x] = \beta_0 + \beta_1 E[x|x] + E[u|x]$$

$$E[y|x] = \beta_0 + \beta_1 x$$

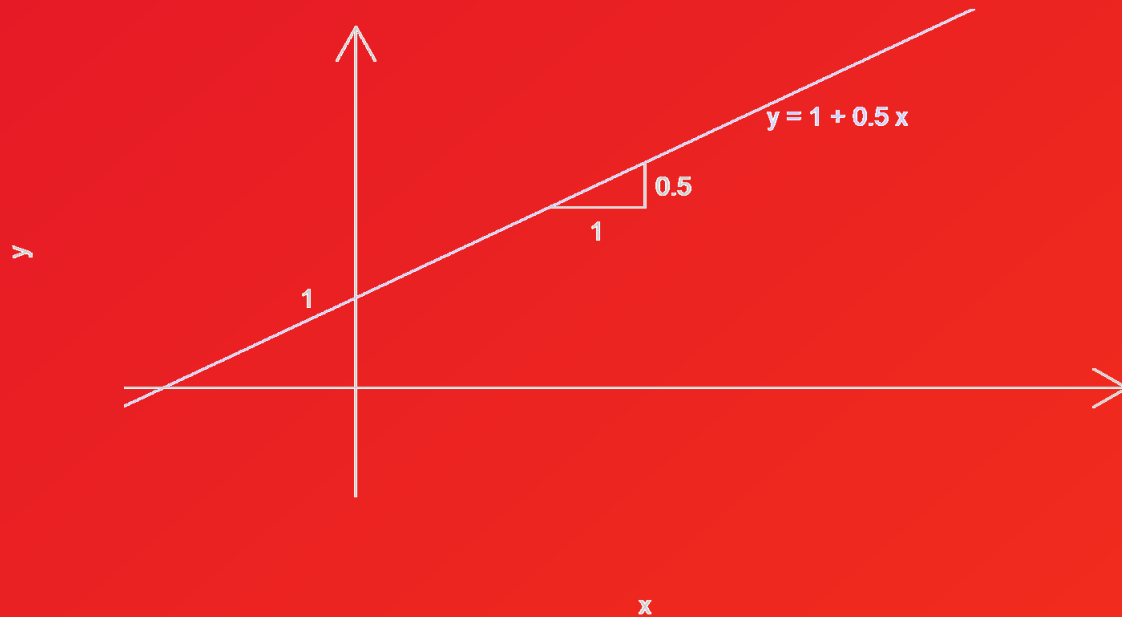
The average of Y , conditional on X , is a linear function of X :

- ✓ For every one-unit increase in X , the expected value of Y changes by the amount β_1 .
- ✓ Given any value of X , the distribution of Y is centred at $E(Y | X)$.

Simple linear regression

$$y = \beta_0 + \beta_1 x$$

$$y = 1 + 0.5x$$



- The intercept, β_0 , represents the value of y when x equals 0.
- The slope, β_1 , represents the increase in y associated with a one-unit increase in x .

Simple linear regression

Bivariate Regression and Covariance

$$COV(X_1, Y_i) = E[(X_i - E[X_i])(Y_i - E[Y_i])]$$

The covariance of a variable with itself is its variance.

$$COV(X_1, X_i) = \sigma_X^2$$

If the expected value of either variable is zero, then the covariance between the two variables is the expected value of their products

$$COV(X_i, Y_i) = E[X_i Y_i]$$

The covariance between linear functions of the variables of interest is given by:

$$W_i = a + bX_i$$

$$Z_i = c + dY_i$$

$$COV(W_i, Z_i) = bdCOV(X_i, Y_i)$$

Simple linear regression

Bivariate Regression and Covariance

The connection between regression and covariance can be observed in a bivariate regression model (with regressor X and dependent variable Y).

The slope and intercept of this regression correspond to the values of β_0 and β_1 that minimise the residual sum of squares (RSS).

$$RSS(a, b) = E[Y_i - \beta_0 - \beta_1 X_i]^2$$

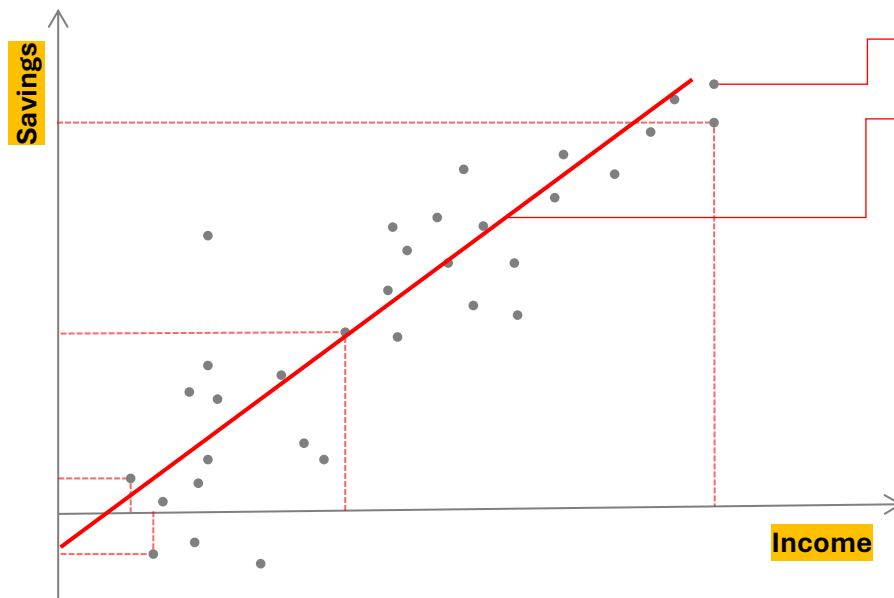
The term RSS refers to a sum of squares because, when performing this minimisation on a particular sample, the expectation is replaced by a sample mean or sample sum.

$$\beta_1 = \frac{COV(Y_i, X_i)}{VAR(X_i)} \quad \beta_0 = E[Y_i] - \beta_1[X_i]$$

An implication of this is that when two variables are not correlated (i.e., have a covariance of zero), the regression of either variable on the other yields a slope coefficient equal to zero.

Simple linear regression

Adjusted Values and Residuals: Ordinary Least Squares (OLS) Estimator



$$(x_i, y_i) : i : 1, \dots, n$$

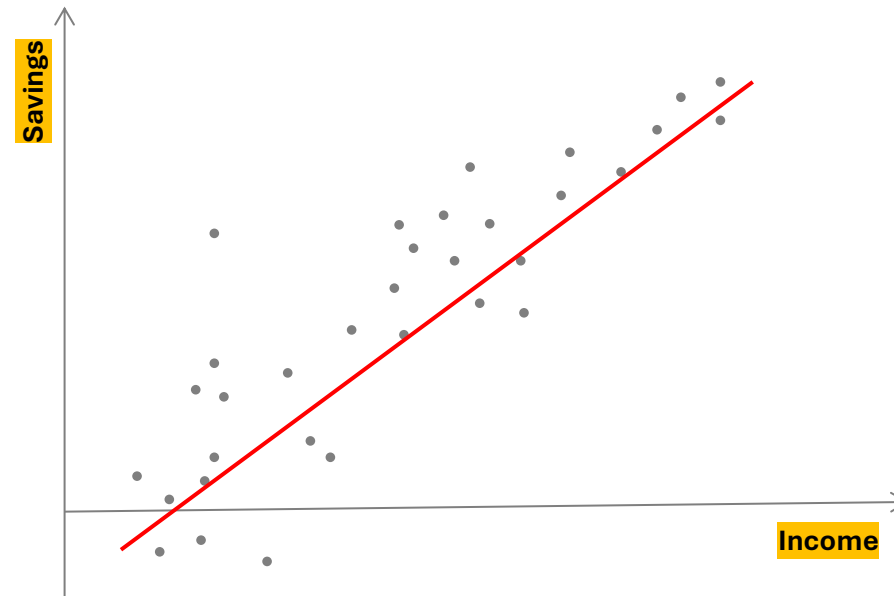
$$y_i = \beta_0 + \beta_1 x_i + u_i$$

- ✓ From a sample, we look to obtain **estimates** of the regression parameters β_0 and β_1 .
- ✓ u_i is the error term for observation i , which aggregates all factors other than X_i that affect Y_i .

How can these sample data be used to obtain estimates of the intercept and slope of the population regression function?

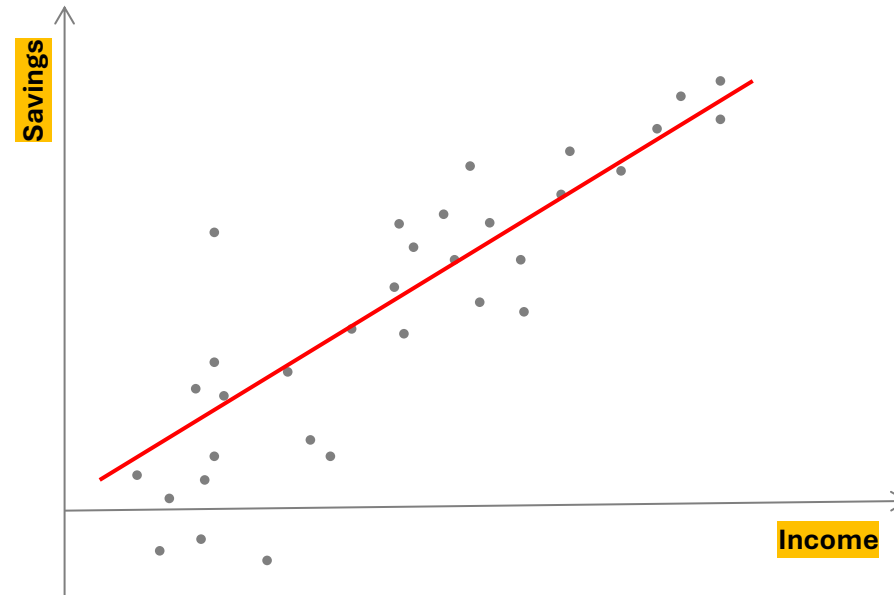
Simple linear regression

Adjusted Values and Residuals: Ordinary Least Squares (OLS) Estimator



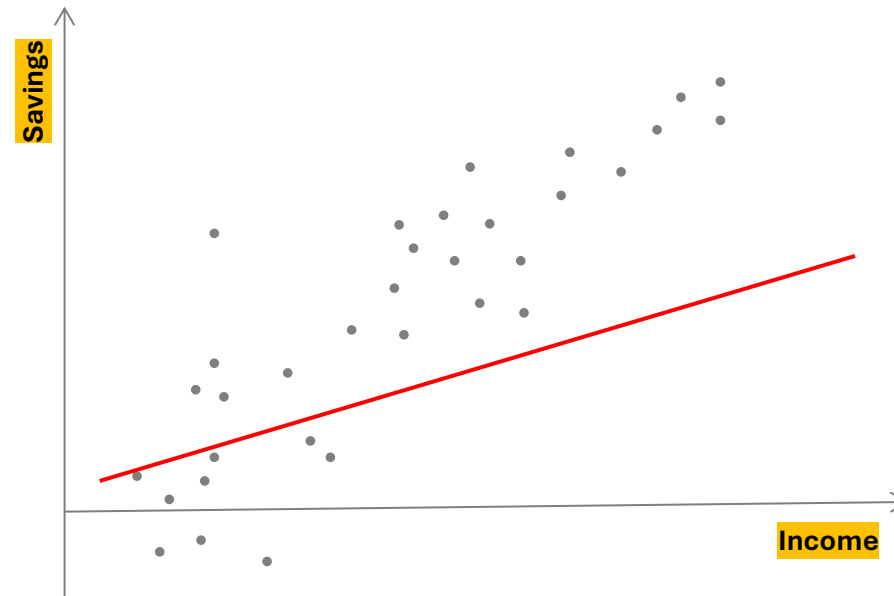
Simple linear regression

Adjusted Values and Residuals: Ordinary Least Squares (OLS) Estimator



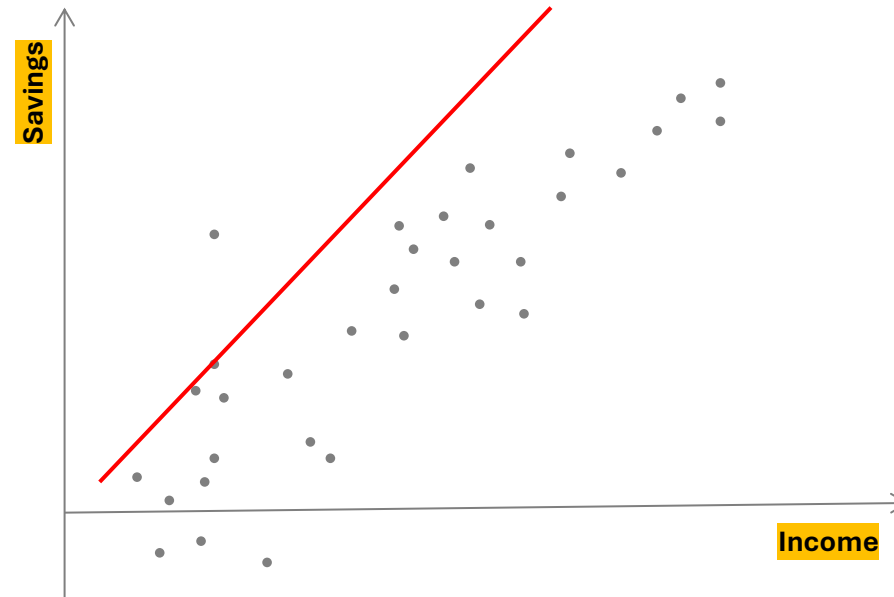
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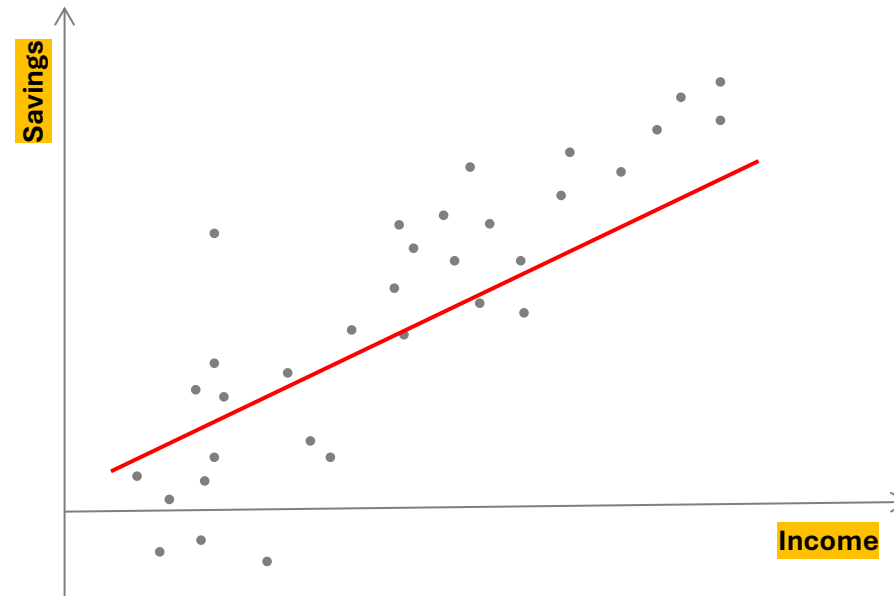
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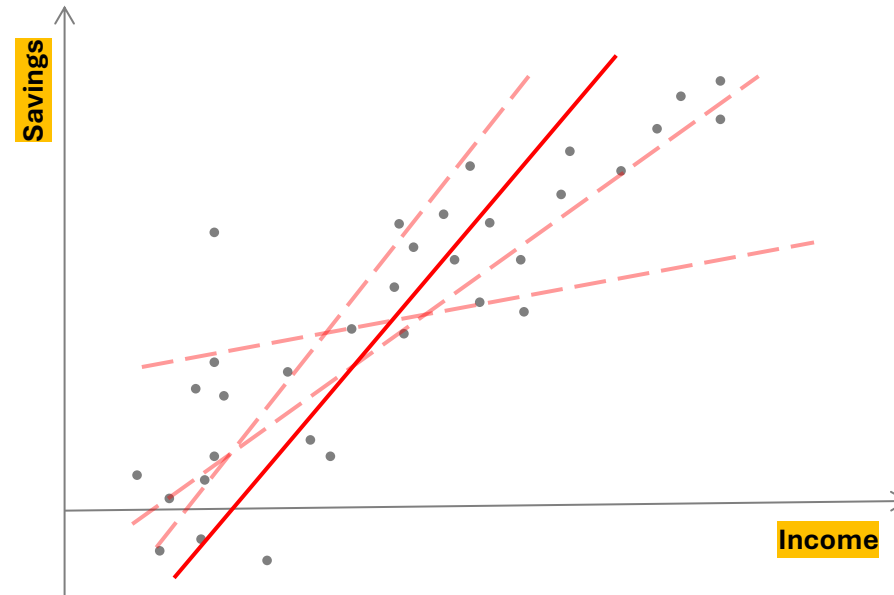
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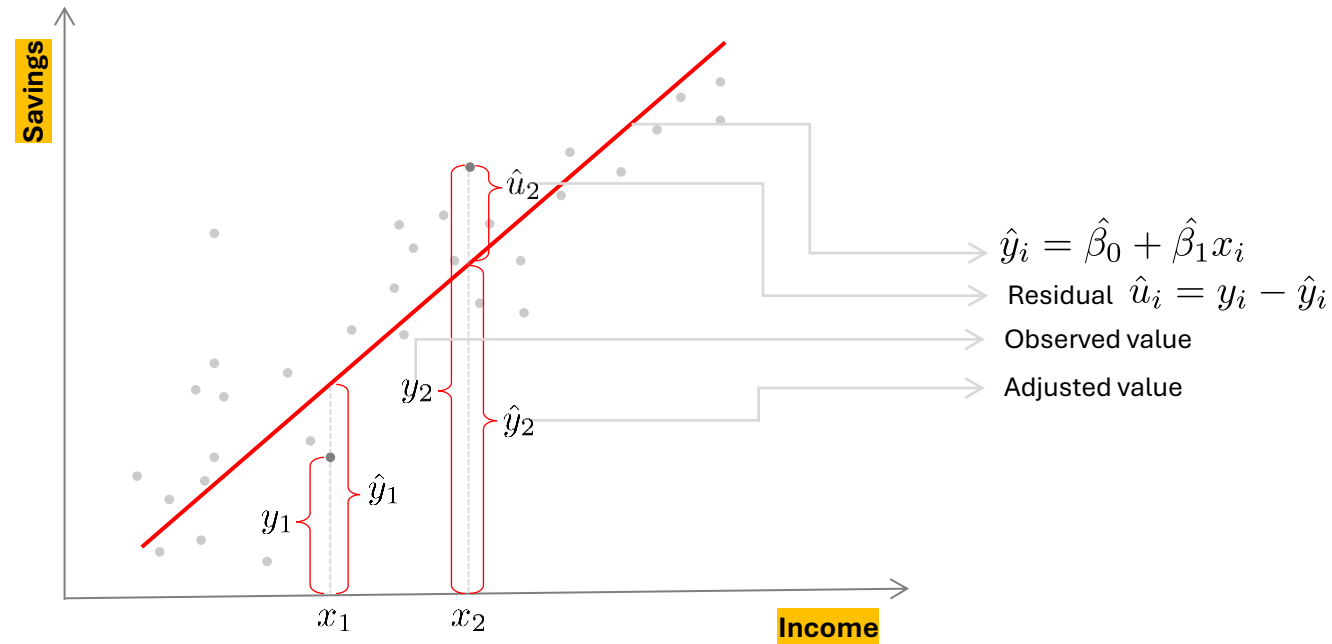
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Simple linear regression

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Simple linear regression

Adjusted Values and Residuals

Ordinary Least Squares (OLS) Estimates

Where do we begin?

$$(x_i, y_i) : i = 1, \dots, n$$

$$y_i = \beta_0 + \beta_1 x_i + u_i$$

Under which conditions?

$$E[u] = 0$$

$$E[u|x] = 0$$

$$\text{Cov}[u, x] = E[xu] = 0$$

If

$$y_i = \beta_0 + \beta_1 x_i + u_i$$

$$u_i = y_i - \beta_0 - \beta_1 x_i$$

Then

$$\text{Cov}[u, x] = E[xu] = E[x(y_i - \beta_0 - \beta_1 x_i)] = 0$$

$$E[u] = E[y_i - \beta_0 - \beta_1 x_i] = 0$$

This introduces **two restrictions on the joint probability distribution** of (x, y) in the population.

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$$y_i = \beta_0 + \beta_1 x_i + u_i$$

Under which conditions?

$$E[u] = 0$$

$$E[u|x] = 0$$

$$\text{Cov}[u, x] = E[xu] = 0$$

What are we looking for?

To choose estimates of β_0 and β_1 that solve the sample counterparts of the equations.

$$E[xu] = E[x(y_i - \beta_0 - \beta_1 x_i)] = 0$$

$$E[u] = E[y_i - \beta_0 - \beta_1 x_i] = 0$$

That is,

$$\frac{1}{n} \sum_{i=1}^n (x_i u_i) = \frac{1}{n} \sum_{i=1}^n (x_i (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)) = 0$$

$$\frac{1}{n} \sum_{i=1}^n (u_i) = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0$$

Simple linear regression

Adjusted Values and Residuals

Ordinary Least Squares (OLS) Estimates

Where do we begin?

$$(x_i, y_i) : i = 1, \dots, n$$

$$y_i = \beta_0 + \beta_1 x_i + u_i$$

Under which conditions?

$$E[u] = 0$$

$$E[u|x] = 0$$

$$\text{Cov}[u, x] = E[xu] = 0$$

What are we looking for? : β_0

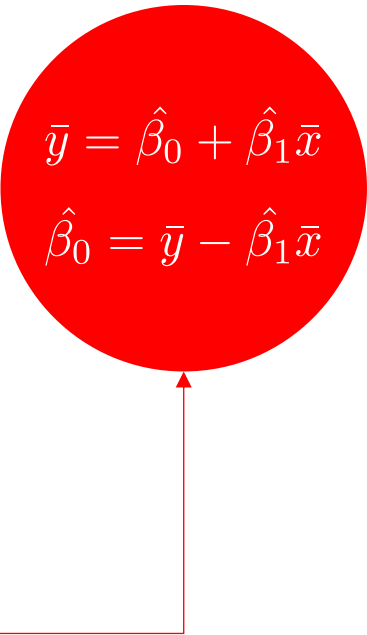
$$\frac{1}{n} \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0$$

$$\frac{1}{n} \left(\sum_{i=1}^n (y_i) - n\hat{\beta}_0 - \hat{\beta}_1 \sum_{i=1}^n x_i \right) = 0$$

$$\frac{1}{n} \sum_{i=1}^n (y_i) - \frac{1}{n} n\hat{\beta}_0 - \hat{\beta}_1 \frac{1}{n} \sum_{i=1}^n x_i = 0$$

$$\frac{1}{n} \sum_{i=1}^n (y_i) - \frac{1}{n} n\hat{\beta}_0 - \hat{\beta}_1 \frac{1}{n} \sum_{i=1}^n x_i = 0$$

$$\bar{y} - \hat{\beta}_0 - \hat{\beta}_1 \bar{x} = 0$$



$$\bar{y} = \hat{\beta}_0 + \hat{\beta}_1 \bar{x}$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

Simple linear regression

Adjusted Values and Residuals

Ordinary Least Squares (OLS) Estimates

Where do we begin?

$$(x_i, y_i) : i = 1, \dots, n$$

$$y_i = \beta_0 + \beta_1 x_i + u_i$$

Under which conditions?

$$E[u] = 0$$

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$$\text{Cov}[u, x] = E[xu] = 0$$

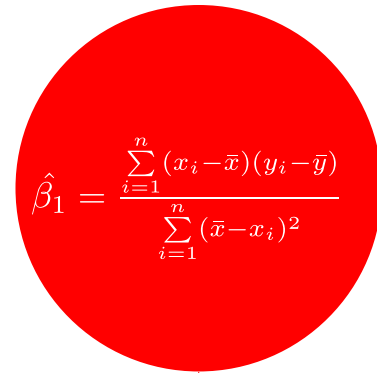
What are we looking for? : β_1

$$\frac{1}{n} \sum_{i=1}^n (x_i (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)) = 0$$

$$\frac{1}{n} \sum_{i=1}^n (x_i (y_i - (\bar{y} - \hat{\beta}_1 \bar{x}) - \hat{\beta}_1 x_i)) = 0$$

$$\frac{1}{n} \sum_{i=1}^n x_i (y_i - \bar{y}) = \hat{\beta}_1 \frac{1}{n} \sum_{i=1}^n x_i (\bar{x} - x_i)$$

$$\frac{1}{n} \sum_{i=1}^n x_i (y_i - \bar{y}) = \hat{\beta}_1 \frac{1}{n} \sum_{i=1}^n x_i (\bar{x} - x_i)$$



$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (\bar{x} - x_i)^2}$$

Simple linear regression

Ordinary Least Squares (OLS) Estimates

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (\bar{x} - x_i)^2}$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

The sample estimation for β_1 is the sample covariance between x and y divided by the sample variance of x .

- If, in the sample, x and y **are positively correlated**, then β_1 is positive
- if x and y are **negatively correlated**, then β_1 is negative.

Simple linear regression

Ordinary Least Squares (OLS) Estimates

How can we study the relationship between years of education and wages?

$$\begin{aligned} \hat{wage} &= \hat{\beta}_0 + \hat{\beta}_1 educ \\ \hat{wage} &= -0.9 + 0.54educ \end{aligned}$$

Variation in the hourly wage corresponding to an **additional year of education**.

$$\Delta \hat{y} = \hat{\beta}_1 \Delta x$$

Each additional year of education, holding all other relevant variables constant, **implies an increase of \$0.54 in the hourly wage**.

If education is zero (no years of schooling), is the wage negative? ⚠️ These may be sample effects!

Simple linear regression

Unbiasedness of OLS estimators



Under LSA 1, 2, 3, and 4, it can be shown that the OLS estimators are unbiased.

$$E[\hat{\beta}_1] = \beta_1 \mid E[\hat{\beta}_0] = \beta_0$$

Unbiasedness is a property of the sampling distributions of β_1 and β_0 ; **it says nothing about the estimates obtained from any particular sample.**

In general, unbiasedness does not hold when any of the four assumptions is not satisfied. This is why it is very important to reflect on the truth of each assumption in the particular application being studied.

Simple linear regression

Unbiasedness of OLS estimators

In general, unbiasedness does not hold when any of the four assumptions is not satisfied. This is why it is very important to reflect on the truth of each assumption in the particular application being studied.

Asumption	What should we ask ourselves 🔥 ?
LSA1: Linearity in the parameters	Can y and x be chosen in such a way that they produce interesting non-linear relationships?
LSA2: Random sampling	In a cross-sectional sample, is the sample representative of the underlying population?
LSA3: Sample variation in x	Within the sample, is there variation in xxx?
LSA4: Zero conditional mean	In the relationship between y and x, might the association actually be driven by unobserved factors that affect y and that happen to be correlated with x?

Simple linear regression

Variance of the OLS estimators

LSA
5

Homoscedasticity

We need to add an assumption in cross-sectional analysis which states that the variance of the unobservable factors u , conditional on x , is constant.

$$\text{Var}(u|x) = \sigma^2$$

The unbiasedness of the regression estimators (which we demonstrated using assumptions RLS1 to RLS4) does not depend on the assumption of homoscedasticity.

- If we assume that u and x are independent, then the distribution of u given x does not depend on x .
- This allows us to see the connection between RLS4 and RLS5.
- Here we have a problem: **the assumption of conditional mean independence can be very strong and difficult to justify.**

LSA4 $E[u|x] = 0$

$$E[u|x] = E[y|x] - E[\beta_0|x] - E[\beta_1 x|x] = 0$$

$$E[y|x] = E[\beta_0|x] + E[\beta_1 x|x]$$

$$E[y|x] = \beta_0 + \beta_1 x$$

LSA5 $\text{Var}[u|x] = \sigma^2$

$$\text{Var}[y|x] = \text{Var}[\beta_0|x] + \text{Var}[\beta_1 x|x] + \text{Var}[u|x]$$

$$\text{Var}[y|x] = \text{Var}[u|x]$$

$$\text{Var}[y|x] = \sigma^2$$

Simple linear regression

Results of the Linear Regression Model

```
Call:
lm(formula = tco2_first ~ pco2_first, data = dat)

Residuals:
    Min       1Q   Median       3Q      Max
-18.8852  -2.5080   0.1891   2.8077  19.2005

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 16.210859   0.359676  45.07  <2e-16 ***
pco2_first   0.188572   0.007886  23.91  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.395 on 1588 degrees of freedom
(186 observations deleted due to missingness)
Multiple R-squared:  0.2647,    Adjusted R-squared:  0.2643
F-statistic: 571.8 on 1 and 1588 DF,  p-value: < 2.2e-16
```

Model structure

$$TCO_2 = 16.21 + 0.19 PCO_2$$

Key Statistics

- **Multiple R-squared:** 0.2647. The model explains approximately 26.47% of the variability in TCO_2
- **Residual standard error:** 4.395. On average, the predicted values of TCO_2 may deviate by ± 4.395 units.
- **F-statistic:** 571.8 (p-value < 2.2e-16). The relationship between PCO_2 and TCO_2 is highly statistically significant.

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- **Estimated Intercept ($\hat{\beta}_0$):** This indicates the predicted level of TCO₂ for an individual with a PCO₂ value of 0. This interpretation is purely mathematical, and such a quantity often has limited practical use.
- **Estimated Slope ($\hat{\beta}_1$):** This can be interpreted as the rate at which the predicted value of TCO₂ increases for each one-unit rise in PCO₂. In this case, it is estimated that TCO₂ increases by approximately 0.189 mmol/L (millimoles per litre) for every 1 mmHg (millimetre of mercury) increase in PCO₂.
- **Standard Error:** This is a measure of the degree of certainty associated with the estimate. If the standard error is large relative to the coefficient, the estimate is less reliable. Many factors can influence the magnitude of the standard error, including the sample size.
- **t-value:** This is simply the estimated coefficient divided by its standard error.
- **Pr(>|t|) or p-value:** This is a measure of statistical significance. Along with the t-statistic, it is linked to hypothesis testing conducted within the model framework.